		is negative. Also, the temperature and hence U increases, making ΔU positive.
19	**C**	**A**: False. Internal energy depends on the temperature of the sphere and number of particles in the sphere. This number is different for the two spheres. **B**: False. The spheres have different surface area, so they do not radiate heat at the same rate. **C**: True. The spheres are at the same temperature, so no net flow of heat between them. **D**: False. Being at the same temperature, the internal energy per atom is the same.
20	**A**	Amount of charge, Q that flowed during the lightning flash = It